Document-QA architecture comparison

Document: **A Suitable Boy Vikram Seth.pdf** | LLM: **gemini-3.5-flash-lite** | run 2026-10-22 11:12 | 30 questions

Architectures

Name	What it is	Source
`closed_book`	no retrieval	baseline
`dense_k5`	dense top-5	naive RAG
`bm25_k5`	BM25 top-5	baseline
`hybrid_k5`	dense + BM25 (RRF) top-5	standard hybrid RAG
`hybrid_rerank_k5`	hybrid + cross-encoder top-5	standard advanced RAG
`hybrid_rerank_k10`	hybrid + cross-encoder top-10	standard advanced RAG, more pages
`hyde`	retrieve with a hypothetical answer	HyDE, Gao et al. 2022 (arXiv 2212.10496)
`crag`	grade the pages, re-search if none is relevant	CRAG, Yan et al. 2024 (arXiv 2401.15884); no web search, no strip refinement
`self_rag`	one answer per page, a critic picks one	Self-RAG, Asai et al. 2023 (arXiv 2310.11511); prompted, not the fine-tuned model
`ircot`	reasoning steps interleaved with retrieval	IRCoT, Trivedi et al. 2022 (arXiv 2212.10509); 3 steps, no few-shot demos
`adaptive_rag`	route by question complexity	Adaptive-RAG, Jeong et al. 2024 (arXiv 2403.14403); LLM prompt instead of the trained classifier
`abrag`	answer, score confidence, read more while it is low	AB-RAG, arXiv 2606.29090 (known from its summary); same confidence signals, one fixed first batch and budget
`snaprag_fixed`	SnapRAG retrieval + verification, AB-RAG off	this work
`snaprag`	SnapRAG: intent-sized batches + AB-RAG	this work

All architectures use the same index, embedder, cross-encoder, answer prompt and LLM. Questions: 3 aggregation, 6 detail, 3 false_premise, 5 inference, 10 multi_hop, 3 numeric (3 unanswerable or false-premise). Scoring is by fixed patterns, not by an LLM judge. Intervals are bootstrap 95% over questions.

Dataset

30 questions written by reading the book; every answerable question lists the facts an answer needs and the gold pages that hold them (57 distinct pages, from page 9 to 428). Multi-hop questions need facts from two or more pages that are not neighbours.

Question type	Easy	Medium	Hard	Total
aggregation	1	1	1	3

Question type	Easy	Medium	Hard	Total
detail	1	5	0	6
false_premise	0	0	3	3
inference	0	2	3	5
multi_hop	0	4	6	10
numeric	1	2	0	3
all	3	14	13	30

Metrics

- **Accuracy**: share of questions answered correctly. An answerable question is correct when the answer contains every required fact (fixed patterns, no LLM judge); an unanswerable or false-premise question is correct when the answer declines or states the correction.
- **Fact score**: mean share of required facts found in the answer (partial credit, answerable questions).
- **Unanswerable handled**: accuracy on the unanswerable and false-premise questions (does the system refuse instead of inventing?).
- **Refused an answerable question**: share of answerable questions on which the answer is a refusal (over-caution).
- **Context recall**: share of required facts that appear in the pages the LLM read. It measures retrieval, independent of the answer.
- **Page hit** and **gold pages read**: whether a gold page was read, and the share of gold pages read.
- **LLM calls, pages read, context tokens, seconds**: cost of one question.
- **Intervals** are percentile bootstrap 95% intervals over questions (2000 resamples, seed 0); comparisons between architectures are paired on the same questions, with an exact two-sided sign test on the questions where they differ.

Run settings

Setting	Value
`LLM_MODEL`	gemini-3.5-flash-lite
`LLM_REASONING_EFFORT`	low
`EMBEDDING_MODEL`	BAAI/bge-small-en-v1.5
`CROSS_ENCODER_MODEL`	cross-encoder/ms-marco-MiniLM-L-6-v2
`PARENT_CHUNK_SIZE`	3000
`CHILD_CHUNK_SIZE`	800
`STOP_CONFIDENCE`	0.8
`MAX_EXTRA_BATCHES`	3

Setting	Value
CONF_W_MODEL	0.5
CONF_W_SIMILARITY	0.25
CONF_W_SPREAD	0.25
temperature	0

Summary

Accuracy ranking: ircot 70%; abrag 70%; snaprag 70%; hybrid_rerank_k5 67%; hybrid_rerank_k10 67%; dense_k5 63%; hybrid_k5 63%; adaptive_rag 63%; snaprag_fixed 63%; crag 57%; bm25_k5 53%; hyde 53%; self_rag 43%; closed_book 7%.

- SnapRAG against the best published method re-implemented here (ircot): 70.0% against 70.0%, +0.0 pp (95% interval -17 to +17 pp; won 3, lost 3, tied 24 of 30 questions): no clear difference.
- Against the extra AB-RAG batches (SnapRAG without them): +6.7 pp (95% interval -13 to +23 pp; won 5, lost 3, tied 22 of 30 questions): no clear difference.
- Against the intent-aware first batch and budget (AB-RAG with one fixed batch for every question): +0.0 pp (95% interval -20 to +20 pp; won 4, lost 4, tied 22 of 30 questions): no clear difference.
- Against a standard hybrid + cross-encoder pipeline: +3.3 pp (95% interval -13 to +20 pp; won 4, lost 3, tied 23 of 30 questions): no clear difference.
- Cost: snaprag makes 3.7 LLM calls per question; hyde 2.0, crag 2.3, self_rag 4.0, ircot 3.3, adaptive_rag 3.0, abrag 3.5.

Answer quality

Accuracy = every required fact is in the answer (unanswerable questions: the answer declines or states the correction).

Architecture	Questions answered	Accuracy, all questions (95% CI)	vs dense_k5	Answerable only	Mean fact score	Unanswerable handled	Refused an answerable question
closed_book	30/30	6.7% (0-17)	-56.7 pp	7.4% (0-19)	14.2%	0.0% (0-0)	0.0%
dense_k5	30/30	63.3% (47-80)	-	59.3% (41-78)	79.6%	100.0% (100-100)	3.7%
bm25_k5	30/30	53.3% (37-70)	-10.0 pp	48.1% (30-67)	71.0%	100.0% (100-100)	11.1%
hybrid_k5	30/30	63.3% (47-80)	+0.0 pp	59.3% (41-78)	75.9%	100.0% (100-100)	7.4%
hybrid_rerank_k5	30/30	66.7% (50-83)	+3.3 pp	63.0% (44-81)	79.0%	100.0% (100-100)	11.1%
hybrid_rerank_k10	30/30	66.7% (50-83)	+3.3 pp	63.0% (44-81)	82.1%	100.0% (100-100)	3.7%
hyde	30/30	53.3% (37-73)	-10.0 pp	48.1% (30-67)	66.0%	100.0% (100-100)	14.8%
crag	30/30	56.7% (40-73)	-6.7 pp	51.9% (33-70)	74.1%	100.0% (100-100)	7.4%
self_rag	30/30	43.3% (27-60)	-20.0 pp	37.0% (19-56)	62.3%	100.0% (100-100)	22.2%
ircot	30/30	**70.0% (53-87)**	+6.7 pp	66.7% (48-85)	85.2%	100.0% (100-100)	3.7%

Architecture	Questions answered	Accuracy, all questions (95% CI)	vs dense_k5	Answerable only	Mean fact score	Unanswerable handled	Refused an answerable question
adaptive_rag	30/30	63.3% (47-80)	+0.0 pp	59.3% (41-78)	77.8%	100.0% (100-100)	11.1%
abrag	30/30	**70.0% (53-87)**	+6.7 pp	66.7% (48-85)	82.7%	100.0% (100-100)	7.4%
snaprag_fixed	30/30	63.3% (47-80)	+0.0 pp	59.3% (41-78)	77.2%	100.0% (100-100)	7.4%
snaprag	30/30	**70.0% (53-87)**	+6.7 pp	66.7% (48-85)	83.3%	100.0% (100-100)	7.4%

Retrieval quality

Context recall = share of required facts present in the pages the LLM read. Page hit = a gold page was read. These are measured on answerable questions only.

Architecture	Context recall (95% CI)	Page hit	Gold pages read	Pages read (mean)	Context (characters)
dense_k5	80.9% (70-91)	88.9% (78-100)	67.3%	5.0	6,632
bm25_k5	72.2% (58-85)	88.9% (78-100)	64.2%	5.0	6,623
hybrid_k5	78.4% (66-90)	88.9% (78-100)	72.2%	5.0	6,667
hybrid_rerank_k5	81.5% (69-93)	88.9% (74-100)	72.8%	5.0	6,690
hybrid_rerank_k10	85.8% (74-95)	92.6% (81-100)	80.2%	10.0	12,340
hyde	67.9% (52-82)	85.2% (70-96)	61.7%	5.0	6,833
crag	77.8% (65-90)	92.6% (81-100)	60.5%	2.4	3,547
self_rag	74.1% (59-87)	85.2% (70-96)	67.3%	3.0	4,475
ircot	**88.9% (79-97)**	96.3% (89-100)	77.8%	4.3	6,374
adaptive_rag	81.5% (69-93)	88.9% (74-100)	74.1%	4.5	6,207
abrag	85.2% (73-95)	92.6% (81-100)	76.5%	7.0	10,002
snaprag_fixed	79.0% (65-90)	92.6% (81-100)	72.2%	5.4	8,016
snaprag	85.8% (74-95)	96.3% (89-100)	77.2%	6.2	9,153

Context recall by question type

Architecture	detail (n=6)	numeric (n=3)	aggregation (n=3)	multi_hop (n=10)	inference (n=5)
dense_k5	100.0%	50.0%	88.9%	73.3%	86.7%
bm25_k5	55.6%	50.0%	66.7%	80.0%	93.3%
hybrid_k5	83.3%	50.0%	100.0%	70.0%	93.3%
hybrid_rerank_k5	100.0%	50.0%	88.9%	71.7%	93.3%
hybrid_rerank_k10	100.0%	66.7%	88.9%	78.3%	93.3%

Architecture	detail (n=6)	numeric (n=3)	aggregation (n=3)	multi_hop (n=10)	inference (n=5)
hyde	72.2%	33.3%	100.0%	60.0%	80.0%
crag	100.0%	50.0%	77.8%	71.7%	80.0%
self_rag	83.3%	50.0%	88.9%	61.7%	93.3%
ircot	100.0%	50.0%	100.0%	88.3%	93.3%
adaptive_rag	100.0%	50.0%	88.9%	71.7%	93.3%
abrag	100.0%	66.7%	88.9%	76.7%	93.3%
snaprag_fixed	83.3%	33.3%	88.9%	80.0%	93.3%
snaprag	83.3%	66.7%	100.0%	85.0%	93.3%

Cost and speed

Context tokens are the characters of the pages the LLM read divided by 4 (an estimate; the answer prompt and the extra calls of the longer pipelines add more). Seconds include the wait for the request-rate limit.

Architecture	LLM calls per question	Accuracy per LLM call	Pages read (mean)	Context tokens (estimate)	Median seconds	95th percentile seconds	Errors
closed_book	1.0	6.7 pp	0.0	0	4.2	144.1	0
dense_k5	1.0	63.3 pp	5.0	1,658	3.5	51.8	0
bm25_k5	1.0	53.3 pp	5.0	1,656	9.7	42.2	0
hybrid_k5	1.0	63.3 pp	5.0	1,667	12.0	25.1	0
hybrid_rerank_k5	1.0	66.7 pp	5.0	1,672	11.3	20.0	0
hybrid_rerank_k10	1.0	66.7 pp	10.0	3,085	12.5	42.6	0
hyde	2.0	26.7 pp	5.0	1,708	33.6	71.6	0
crag	2.3	25.0 pp	2.4	887	52.4	61.9	0
self_rag	4.0	10.8 pp	3.0	1,119	67.2	233.4	0
ircot	3.3	21.4 pp	4.3	1,594	53.1	110.0	0
adaptive_rag	3.0	21.1 pp	4.5	1,552	56.0	117.0	0
abrag	3.5	20.2 pp	7.0	2,501	67.2	284.7	0
snaprag_fixed	3.1	20.7 pp	5.4	2,004	35.1	57.9	0
snaprag	3.7	18.9 pp	6.2	2,288	59.5	107.8	0

Self-verification (architectures with a confidence check)

Verified = the checker found the answer supported by the retrieved text. A useful check separates answers that are right from answers that are wrong: accuracy should be higher among the verified ones.

Architecture	Mean confidence	Answers verified	Accuracy when verified	Accuracy when not verified
abrag	0.86	100.0%	70.0% (53-87)	-
snaprag_fixed	0.86	96.7%	65.5% (48-83)	0.0% (0-0)
snaprag	0.88	100.0%	70.0% (53-87)	-

adaptive_rag routed 0 questions to no retrieval, 17 to single-step retrieval and 13 to multi-step retrieval.

Accuracy by question type

Architecture	detail (n=6)	numeric (n=3)	aggregation (n=3)	multi_hop (n=10)	inference (n=5)	false_premise (n=3)
closed_book	0.0%	33.3%	0.0%	0.0%	20.0%	0.0%
dense_k5	100.0%	33.3%	66.7%	40.0%	60.0%	100.0%
bm25_k5	50.0%	33.3%	66.7%	30.0%	80.0%	100.0%
hybrid_k5	83.3%	33.3%	100.0%	30.0%	80.0%	100.0%
hybrid_rerank_k5	100.0%	33.3%	66.7%	40.0%	80.0%	100.0%
hybrid_rerank_k10	100.0%	66.7%	66.7%	30.0%	80.0%	100.0%
hyde	66.7%	33.3%	100.0%	30.0%	40.0%	100.0%
crag	100.0%	33.3%	66.7%	30.0%	40.0%	100.0%
self_rag	50.0%	33.3%	66.7%	10.0%	60.0%	100.0%
ircot	100.0%	33.3%	100.0%	40.0%	80.0%	100.0%
adaptive_rag	100.0%	33.3%	66.7%	30.0%	80.0%	100.0%
abrag	100.0%	66.7%	66.7%	40.0%	80.0%	100.0%
snaprag_fixed	83.3%	33.3%	66.7%	50.0%	60.0%	100.0%
snaprag	83.3%	66.7%	100.0%	50.0%	60.0%	100.0%

Accuracy by difficulty

Architecture	easy (n=3)	medium (n=14)	hard (n=13)
closed_book	33.3%	7.1%	0.0%
dense_k5	100.0%	64.3%	53.8%
bm25_k5	100.0%	35.7%	61.5%

Architecture	easy (n=3)	medium (n=14)	hard (n=13)
hybrid_k5	100.0%	57.1%	61.5%
hybrid_rerank_k5	100.0%	64.3%	61.5%
hybrid_rerank_k10	100.0%	71.4%	53.8%
hyde	100.0%	57.1%	38.5%
crag	100.0%	57.1%	46.2%
self_rag	100.0%	35.7%	38.5%
ircot	100.0%	71.4%	61.5%
adaptive_rag	100.0%	64.3%	53.8%
abrag	100.0%	71.4%	61.5%
snaprag_fixed	66.7%	64.3%	61.5%
snaprag	100.0%	78.6%	53.8%

snaprag against every other architecture (paired, same questions)

Each cell: mean difference in percentage points (95% bootstrap interval); questions won / lost / tied; 'better' or 'worse' only when the interval excludes zero. The sign test is an exact two-sided binomial test on the questions where the two architectures differ.

Compared with	Accuracy difference	Fact-score difference	Context-recall difference
closed_book	+63.3 pp (+47 to +80); 19/0/11; sign test p=0.000; better	+72.2 pp (+58 to +86); 26/0/4; better	+85.8 pp (+74 to +95); 25/0/2; better
dense_k5	+6.7 pp (-10 to +27); 5/3/22; sign test p=0.727; no clear difference	+3.3 pp (-8 to +15); 5/3/22; no clear difference	+4.9 pp (-9 to +19); 6/3/18; no clear difference
bm25_k5	+16.7 pp (+0 to +37); 7/2/21; sign test p=0.180; no clear difference	+11.1 pp (+1 to +23); 7/2/21; better	+13.6 pp (+2 to +27); 6/1/20; better
hybrid_k5	+6.7 pp (-10 to +23); 4/2/24; sign test p=0.688; no clear difference	+6.7 pp (-2 to +17); 5/2/23; no clear difference	+7.4 pp (-2 to +19); 4/1/22; no clear difference
hybrid_rerank_k5	+3.3 pp (-13 to +20); 4/3/23; sign test p=1.000; no clear difference	+3.9 pp (-8 to +16); 5/3/22; no clear difference	+4.3 pp (-9 to +17); 5/3/19; no clear difference
hybrid_rerank_k10	+3.3 pp (-13 to +20); 4/3/23; sign test p=1.000; no clear difference	+1.1 pp (-11 to +12); 4/3/23; no clear difference	+0.0 pp (-12 to +12); 3/3/21; no clear difference
hyde	+16.7 pp (+0 to +33); 6/1/23; sign test p=0.125; no clear difference	+15.6 pp (+5 to +26); 9/1/20; better	+17.9 pp (+7 to +30); 8/1/18; better
crag	+13.3 pp (-7 to +33); 7/3/20; sign test p=0.344; no clear difference	+8.3 pp (-4 to +21); 8/3/19; no clear difference	+8.0 pp (-6 to +20); 8/3/16; no clear difference

Compared with	Accuracy difference	Fact-score difference	Context-recall difference
`self_rag`	+26.7 pp (+10 to +47); 9/1/20; sign test p=0.021; better	+18.9 pp (+8 to +32); 10/1/19; better	+11.7 pp (+0 to +25); 6/2/19; no clear difference
`ircot`	+0.0 pp (-17 to +17); 3/3/24; sign test p=1.000; no clear difference	-1.7 pp (-12 to +7); 3/3/24; no clear difference	-3.1 pp (-14 to +6); 2/3/22; no clear difference
`adaptive_rag`	+6.7 pp (-13 to +23); 5/3/22; sign test p=0.727; no clear difference	+5.0 pp (-7 to +17); 6/3/21; no clear difference	+4.3 pp (-9 to +17); 5/3/19; no clear difference
`abrag`	+0.0 pp (-20 to +20); 4/4/22; sign test p=1.000; no clear difference	+0.6 pp (-11 to +11); 5/4/21; no clear difference	+0.6 pp (-12 to +12); 4/3/20; no clear difference
`snaprag_fixed`	+6.7 pp (-13 to +23); 5/3/22; sign test p=0.727; no clear difference	+5.6 pp (-4 to +16); 5/3/22; no clear difference	+6.8 pp (-2 to +18); 4/1/22; no clear difference

Questions `snaprag` got wrong (9)

- **q06** (multi_hop, medium): Where does Arun Mehra work, what is his brother Varun studying, and where do the two of them live?

expected: Arun is an executive at Bentsen & Pryce in Calcutta; Varun studies mathematics at Calcutta University and lives with Arun and Meenakshi in Calcutta.

answered: - Arun Mehra's workplace: Not found in the retrieved text: the specific place where Arun Mehra works (though he goes to an office, PDF p. 381). - Varun is studying mathematics at Calcutta University (PDF p. 22). - They live in Arun and Meenakshi's small ground

- **q08** (inference, hard): What does Arun look down on about the Kapoors and the wedding celebrations?

expected: He finds them provincial and crass: the rustic relatives and state politicians, the coloured lights in the hedge, the pomegranate juice instead of alcohol, and Maan running a cloth shop in Banaras.

answered: The text states: - Arun considers the Kapoors to be merely "all very well for Brahmpur society" and compares them unfavourably to his in-laws, noting that Pran's brother ran "a cloth shop in Banaras" (PDF p. 26). - He looks down on the wedding celebrations as

- **q13** (detail, medium): Who saved the convoy Kedarnath was travelling in when it was attacked near the Ravi Bridge?

expected: A Gurkha regiment coming from the other side opened fire and the attackers fled.

answered: Insufficient information found.

- **q33** (multi_hop, hard): What orders did Home Minister L.N. Agarwal give District Magistrate Krishan Dayal about Misri Mandi, and what practical objection did the DM raise?

expected: He told him to guard the entrances of the Govind Shoe Mart and disperse pickets, reading Section 144 and firing if necessary; the DM objected that police were tied up at the Shiva Temple site.

answered: **The text states:** Home Minister L.N. Agarwal ordered District Magistrate Krishan Dayal to guard the entrance of the Govind Shoe Mart, disperse any pickets forthwith, read out Section 144 of the Criminal Procedure Code if necessary, and then fire if the crow

- **q36** (multi_hop, hard): How was the police action at Baitar House stopped, and who carried the appeal to the Chief Minister?

expected: Zainab wrote a letter that Murtaza Ali and little Hassan carried to the Chief Minister, who phoned Agarwal and ordered him to call it off and make the police leave.

answered: **The text states:** * Murtaza Ali carried the appeal to the Chief Minister via a letter ([Source: A Suitable Boy Vikram Seth.pdf, PDF page 241]). * The Chief Minister stopped the action by calling the Home Minister (Agarwal) and ordering him to "call it off a

- **q41** (numeric, medium): How many people did Begum Abida Khan say were being dispossessed by the bill?

expected: Eight lakh people.

answered: Not found in the retrieved text: the number of people being dispossessed by the bill (though the text notes that hundreds of people depend on the establishments of larger landlords, PDF p. 244).

- **q46** (inference, hard): Why was Ustad Majeed Khan upset about the Shiva Temple, and what did he ask Veena to tell her father?

expected: The temple being built in Chowk near his house would let the conch compete with the muezzin's call to prayer; he asked her to tell her father, the Minister, to help protect their religion.

answered: Based on the provided context: * **Why Ustad Majeed Khan was upset about the Shiva Temple:** The text states he was concerned because once bhajans start being sung from the temple while they are at prayer, "it can only end badly" (PDF p. 266). * **What he aske

- **q48** (multi_hop, hard): Why is Ishaq Khan trying to get his brother-in-law moved, and how do artists' grades matter?

expected: He wants his sister's husband, a sarangi player and staff artist, moved from AIR Lucknow to AIR Brahmpur; the Station Director would not swap a B-plus artist like Rafiq for a B-grade one.

answered: ### Reasons for the Brother-in-Law's Move * **Support for Younger Brothers:** After their father's death, Ishaq Khan and his sister had to support their three young brothers (PDF p. 272, 356). * **Need for Assistance:** Because of this shared family responsibi

- **q58** (multi_hop, hard): What book did Dr Ila Chattopadhyay write, and what did she think of the ghazal singer's performance she attended in Brahmpur?

expected: She wrote 'Metaphysical Causality', a book about Donne; she found the ghazal singer (Saeeda something) dreadful and the self-pitying verse insipid.

answered: The text states that Dr. Ila Chattopadhyay discussed a book about Donne called *Metaphysical Causality* (PDF p. 426), referring to it as "one of my old books" (PDF p. 425) and calling it "a very stupid book" (PDF p. 426). Not found in the retrieved text: what

Notes

- One document, one LLM and 30 questions written by reading the book: the intervals show how little this many questions can separate close systems. Treat differences smaller than the interval as noise.
- The published methods are simplified re-implementations from the papers' descriptions, run on the same small LLM as everything else; they are not the authors' trained models or code, so a result here does not reproduce a paper's own numbers.
- The book is a scanned novel; some pages have broken text, which lowers every architecture equally.
- `closed_book` shows what the LLM already knows about this famous novel; a retrieval system should beat it on details and on the unanswerable questions.
- SnapRAG's AB-RAG weights and threshold are uncalibrated defaults.

Document-QA architecture comparison

Document: **A Suitable Boy Vikram Seth.pdf** | LLM: **none (retrieval only)** | run 2026-10-22 10:45 | 75 questions

Architectures

Name	What it is	Source
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`bm25_k5`	BM25 top-5	baseline
`hybrid_k5`	dense + BM25 (RRF) top-5	standard hybrid RAG
`hybrid_rerank_k5`	hybrid + cross-encoder top-5	standard advanced RAG
`hybrid_rerank_k10`	hybrid + cross-encoder top-10	standard advanced RAG, more pages
`snaprag_fixed`	SnapRAG retrieval + verification, AB-RAG off	this work

All architectures use the same index, embedder and cross-encoder; no LLM answers are generated, and SnapRAG's planner and HyDE run their offline fallbacks. Questions: 6 aggregation, 16 detail, 7 false_premise, 14 inference, 23 multi_hop, 7 numeric, 2 unanswerable (9 unanswerable or false-premise). Scoring is by fixed patterns, not by an LLM judge. Intervals are bootstrap 95% over questions.

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inference	0	5	9	14
multi_hop	0	10	13	23
numeric	2	4	1	7
unanswerable	0	0	2	2
all	4	36	35	75

Metrics

- **Accuracy**: share of questions answered correctly. An answerable question is correct when the answer contains every required fact (fixed patterns, no LLM judge); an unanswerable or false-premise question is correct when the answer declines or states the correction.
- **Fact score**: mean share of required facts found in the answer (partial credit, answerable questions).
- **Unanswerable handled**: accuracy on the unanswerable and false-premise questions (does the system refuse instead of inventing?).
- **Refused an answerable question**: share of answerable questions on which the answer is a refusal (over-caution).
- **Context recall**: share of required facts that appear in the pages the LLM read. It measures retrieval, independent of the answer.
- **Page hit** and **gold pages read**: whether a gold page was read, and the share of gold pages read.
- **LLM calls, pages read, context tokens, seconds**: cost of one question.
- **Intervals** are percentile bootstrap 95% intervals over questions (2000 resamples, seed 0); comparisons between architectures are paired on the same questions, with an exact two-sided sign test on the questions where they differ.

Run settings

Setting	Value
LLM_MODEL	gemini-3.5-flash-lite
LLM_REASONING_EFFORT	low
EMBEDDING_MODEL	BAAI/bge-small-en-v1.5
CROSS_ENCODER_MODEL	cross-encoder/ms-marco-MiniLM-L-6-v2
PARENT_CHUNK_SIZE	3000
CHILD_CHUNK_SIZE	800
STOP_CONFIDENCE	0.8
MAX_EXTRA_BATCHES	3
CONF_W_MODEL	0.5
CONF_W_SIMILARITY	0.25
CONF_W_SPREAD	0.25
temperature	0

Retrieval quality

Context recall = share of required facts present in the pages the LLM read. Page hit = a gold page was read. These are measured on answerable questions only.

Architecture	Context recall (95% CI)	Page hit	Gold pages read	Pages read (mean)	Context (characters)
dense_k5	81.2% (74-88)	86.4% (77-94)	69.1%	5.0	6,432

Architecture	Context recall (95% CI)	Page hit	Gold pages read	Pages read (mean)	Context (characters)
bm25_k5	80.4% (72-88)	89.4% (82-95)	68.8%	5.0	6,429
hybrid_k5	84.0% (77-91)	90.9% (83-97)	73.4%	5.0	6,314
hybrid_rerank_k5	87.4% (81-93)	95.5% (89-100)	77.8%	5.0	6,533
hybrid_rerank_k10	**91.4% (86-96)**	97.0% (92-100)	85.2%	10.0	12,056
snaprag_fixed	90.9% (85-96)	95.5% (89-100)	81.6%	5.0	7,134

Context recall by question type

Architecture	detail (n=16)	numeric (n=7)	aggregation (n=6)	multi_hop (n=23)	inference (n=14)
dense_k5	90.6%	78.6%	84.7%	71.0%	86.9%
bm25_k5	83.3%	54.8%	68.1%	83.3%	90.5%
hybrid_k5	87.5%	69.0%	90.3%	79.0%	92.9%
hybrid_rerank_k5	100.0%	69.0%	88.9%	79.0%	95.2%
hybrid_rerank_k10	100.0%	85.7%	94.4%	84.1%	95.2%
snaprag_fixed	100.0%	85.7%	88.9%	84.1%	95.2%

snaprag_fixed against every other architecture (paired, same questions)

Each cell: mean difference in percentage points (95% bootstrap interval); questions won / lost / tied; 'better' or 'worse' only when the interval excludes zero. The sign test is an exact two-sided binomial test on the questions where the two architectures differ.

Compared with	Accuracy difference	Fact-score difference	Context-recall difference
dense_k5	-	-	+9.7 pp (+4 to +16); 13/1/52; better
bm25_k5	-	-	+10.5 pp (+3 to +19); 12/3/51; better
hybrid_k5	-	-	+6.9 pp (+1 to +13); 10/2/54; better
hybrid_rerank_k5	-	-	+3.5 pp (+1 to +7); 4/0/62; better
hybrid_rerank_k10	-	-	-0.5 pp (-4 to +3); 1/2/63; no clear difference

Notes

- One document, one LLM and 75 questions written by reading the book: the intervals show how little this many questions can separate close systems. Treat differences smaller than the interval as noise.
- The published methods are simplified re-implementations from the papers' descriptions, run on the same small LLM as everything else; they are not the authors' trained models or code, so a result here does not reproduce a paper's own numbers.

- The book is a scanned novel; some pages have broken text, which lowers every architecture equally.
- `closed_book` shows what the LLM already knows about this famous novel; a retrieval system should beat it on details and on the unanswerable questions.
- SnapRAG's AB-RAG weights and threshold are uncalibrated defaults.